

System memory report with top 10 consuming processes

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Date: 2026-04-22

Displays system memory usage in human-readable format and lists the top 10 memory-consuming processes by percentage of memory used and resident set size in MB.

Language

bash

Category

monitoring

Command

```
echo -e "Memory Report:\n$(free -h)\n\nTop 10 Memory\nConsuming Processes (%MEM and RSS in MB):\n$(ps -eo pid,\ncomm,%mem,rss --sort=-%mem | awk 'NR==1 {print $0 " RSS(\nMB)"}; NR>1 {printf "%s %s %s %.2f\n", $1, $2, $3, $4\n/1024}' | head -n 11)"
```

Explanation

The command uses 'free -h' to show memory statistics in human-readable units (e.g., MB, GB). It then uses 'ps' to list all processes sorted by memory usage (%MEM) in descending order, showing the top 10 processes including their RSS (Resident Set Size) converted to MB. The awk command formats the output to display PID, COMMAND, %MEM, and RSS(MB) with proper headers. The output is formatted with clear section headers and spacing for readability.

Tags

memory, system-monitoring, top-processes, ram, ps, free

Dependencies

bash, coreutils, procps-ng

Arguments

None

Examples

1. `echo -e "Memory Report:\n$(free -h)\n\nTop 10 Memory Consuming Processes (%MEM and RSS in MB):\n$(ps -eo pid,comm,%mem,rss --sort=-%mem | awk 'NR==1 {print $0 " RSS(MB)"}; NR>1 {printf "%s %s %s %.2f\n", $1, $2, $3, $4/1024}' | head -n 11)"` - Run the memory report one-liner directly in the shell
2. `alias memreport='echo -e "Memory Report:\n$(free -h)\n\nTop 10 Memory Consuming Processes (%MEM and RSS in MB):\n$(ps -eo pid,comm,%mem,rss --sort=-%mem | awk 'NR==1 {print $0 " RSS(MB)"}; NR>1 {printf "%s %s %s %.2f\n", $1, $2, $3, $4/1024}' | head -n 11)''` && `memreport` - Define as an alias and then execute it

Output

Memory Report:				
	total	used	free	shared
	buff/cache	available		
Mem:	7.7G	2.3G	3.1G	256M
	2.3G	5.0G		
Swap:	2.0G	0B	2.0G	
Top 10 Memory Consuming Processes (%MEM and RSS in MB):				
PID	COMMAND	%MEM	RSS(MB)	
1234	firefox	15.2	1175.04	
5678	chrome	12.8	991.42	
9012	code	8.5	657.92	
3456	thunderbird	6.2	479.74	
7890	slack	4.1	317.12	
1122	python3	3.8	293.76	
3344	java	3.5	270.85	
5566	docker	2.9	224.26	
7788	Xorg	2.5	193.28	
9900	gnome-shell	2.2	170.24	

Notes

- The %MEM column shows the percentage of available physical memory used by the process.

- The RSS(MB) column shows the Resident Set Size in megabytes (non-swapped physical memory used).
- The header line is included in the top 10 output for clarity.
- The awk command converts RSS from KB to MB by dividing by 1024 and formats to 2 decimal places.

Warnings

None

See Also

- [disk-usage-summary](#)
- [cpu-top-processes](#)

Status

reviewed

Safety

safe

Shell

bash

Platforms

linux, gnu-linux, freebsd, openbsd, netbsd

Created At

2026-04-22

Updated At

2026-04-22